

Changwei Yao

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EDUCATION

Carnegie Mellon University
M.S. in Information Systems
Advisor: Marios Savvides

Pittsburgh, PA
Aug. 2024 - May. 2026 (expected)

Hunan University
B.E. in Information Security (1/41)

Changsha, China
Sep. 2019 - Jun. 2023

SKILLS & COURSEWORK

Operating System	Linux (Ubuntu and Arch), Windows
Programming Languages	Python, C++, MATLAB, JavaScript, $\LaTeX$
Physical Robots	LBR iiwa7, ARX, UR5, myCobot
Common Libraries	PyTorch, DM-Control, Isaccgym, Numpy/Scipy, Pandas
Cousework	Linear Algebra, Calculus, Probability, Robot Learning, Learning for 3D Vision

PUBLICATION

- S. Yao, Z. Yan, Y. Li, Q. Qiao, **Changwei Yao**, X. Ying, X. Liu, W. Jin "TOXRobo: Cross-Modal Chain-of-Thought Trojan Exploitation for Malfunctioning Embodied AI Robots." *2026 Conference on Computer Vision and Pattern Recognition (CVPR)*. (under review)
- Y. Lu, P. Zheng, Y. Li, **Changwei Yao**, Y. Xue, J. Luo "Neural Physical Watermarking via Differentiable Print-Capture Simulation." *2026 Conference on Computer Vision and Pattern Recognition (CVPR)*. (under review)
- **Changwei Yao**, X. Liu, C. Li, M. Savvides. "Reward Evolution with Graph-of-Thoughts: A Bi-Level Language Model Framework for Reinforcement Learning." *2026 IEEE International Conference on Robotics and Automatio (ICRA)*. (under review) [\[paper\]](#)
- Q. Wang*, **Changwei Yao***, X. Liu*, Z. Ouyang, J. Liu, H. Lu, W. Zhang. "MinInter: Minimizing Trajectory Interpolation During Data Augmentation for Imitation Learning." *2026 IEEE International Conference on Robotics and Automatio (ICRA)*. (under review)
- S. Zhao, G. Chen, H. Zhu, Z. Li, **Changwei Yao**, N. Zantout, S. Kim, W. Wang, J. Zhang, S. Scherer. "SuperMap: Spatio Temporal Semantic SLAM Enabling Robots to Understand Evolving World in Real Time." [\[demo\]](#)
- Z. Xu, W. Jin, **Changwei Yao**, X. Liu, S. Ma, Y. Liu, Z. Qin, I. Vakilinia, and D. Liu, "EM-Rhythm: An authentication method for heterogeneous IoT devices," *ACM Transactions on Sensor Networks*, vol. 20, no. 6, pp. 1–21, 2024. [\[paper\]](#)
- M. Yang, **Changwei Yao**^{*#}, and S. Yan, "Age Estimation Based on Graph Convolutional Networks and Multi-head Attention Mechanisms," *2023 IEEE 6th International Conference on Information Systems and Computer Aided Education (ICISCAE)*, pp. 1124–1129, 2023.

EXPERIENCE

CyLab Biometrics Center
Part-time Research Assistant. Supervised by Prof. Marios Savvides

Sep. 2024 - Aug. 2025
CMU, PA

- Developed a chat AI agent for navigation at Dubai Airport.
- Enabled LLM to decompose manipulation tasks with **Graph of Thoughts**, helping LLM design more effective reward functions for RL agent in simulation.
- Finding a way to align the **affordance map** and language with the first observed frame of the task. Exploring better methods to enable the **low-level controller** to guide actions more effectively within the VLA framework.

Ji's Lab at Robotics Institute
Part-time Research Assistant. Supervised by Prof. Ji Zhang

Sep. 2024 - Mar. 2025
CMU, PA

- Introduced a unified **scene graph** to track objects over time, which allows a robot to build an abstract metric-semantic representation.
- Leverage open-vocabulary 3D semantics network to support the detection and tracking of both short-term and long-term **dynamic objects**.

Control & Learning for Robotics and Autonomy Lab

Jun. 2023 - Jun. 2024

Full-time Research Assistant. Supervised by Prof. Wei Zhang

SUSTech, China

- Implemented a **distillation** method for Diffusion Policy, enhancing inference speed for efficient, low-latency visuomotor control in resource-constrained robots.
- Explored a novel **data augmentation** method using model-free optimization for imitation learning.

IoT Sensing and Security Lab

Oct. 2021 - Aug. 2022

Part-time Research Assistant. Supervised by Prof. Wenqiang Jin

HNU, China

- Researched machine learning, signal processing, and their application in IoT security.
- Designed a system for heterogeneous IoT device security using electromagnetic signals.

PROJECTS

Deep Reinforcement Learning for Minecraft

Apr. 2023 - May. 2023

Personal Project.

Changsha, China

- **MineRL Integration:** Encapsulated the Minecraft environment within the MineRL library.
- **AI Agent Training:** Trained a tree-chopping AI agent using Deep Q-Learning to learn from demonstrations and achieve automated task execution.

Computer Vision Somatic Analysis Method System

Mar. 2022 - May. 2022

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Changsha, China

- **Software Development:** Developed an IoT software using Heartbeat technology for resource takeover, emphasizing multi-threading and dynamics.
- **Diverse Functionality:** Implemented numerous functions, including Analysis Method View & Management, Posture Analysis Matching, etc.

HONORS & AWARDS

Second Place Award

National College Student Information Security Competition

Meritorious Winner

COMAP's Mathematical Contest in Modeling

Second Place Award

MathorCup College Mathematical Modeling Contest

Merit Award

Electronic Systems Engineering Skills Competition